PISTA

Preventive Investigation System for Transaction Auditing

Pista(chio) provides a hard shell for your nuts to ensure they stays green inside

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The objective

Provide a system that:

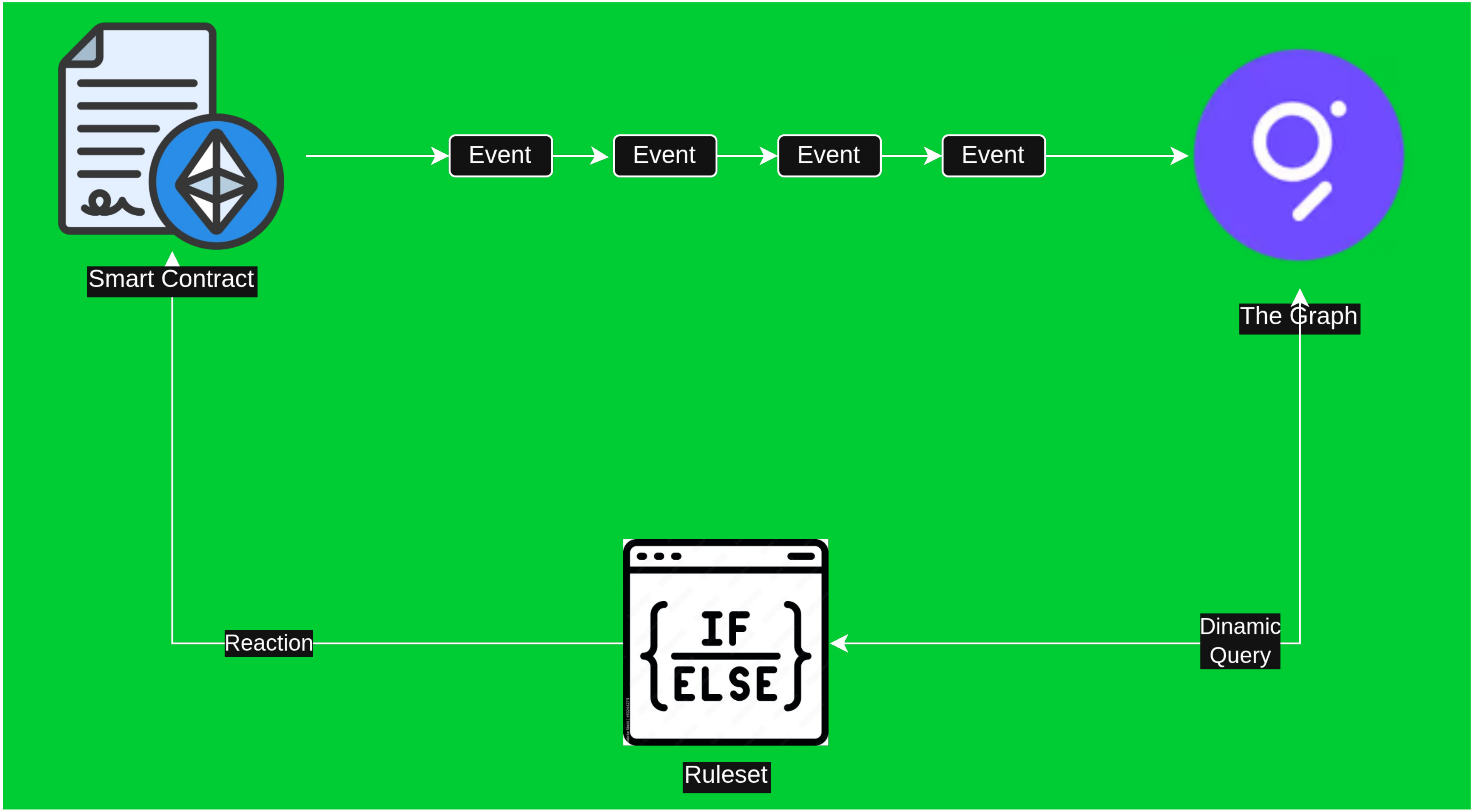
- Detect **unusual** usage patterns while they happen
- Allows smart contracts to **react** to new threads
- Stop **novel** attacks before they spread too far
- Fine tune the response with **complex** logic like
  - bull/bear market
  - cross chain events
  - broad market status
- With **flexible** logic that changes over time

The problem

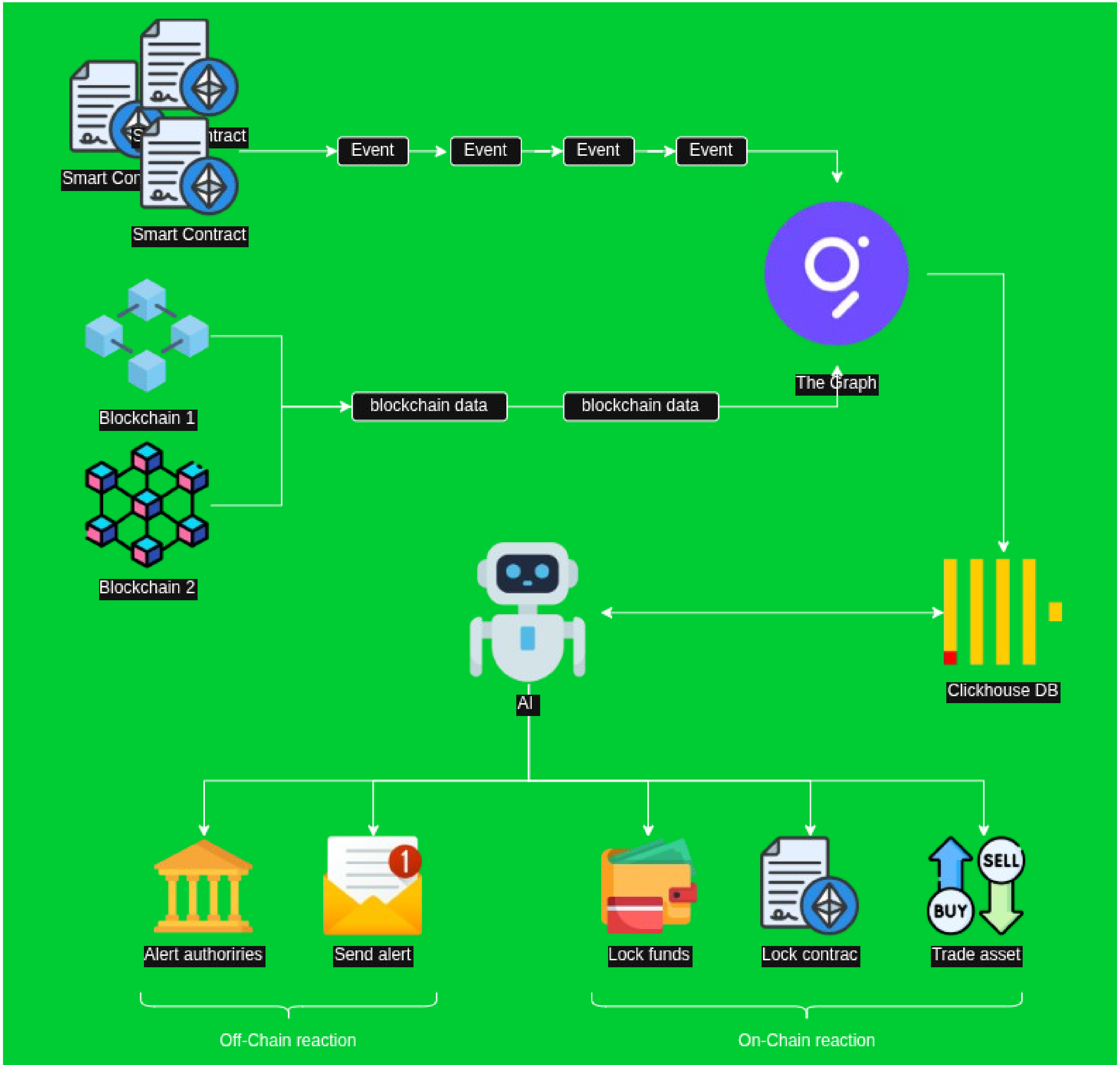
- Encoding **complex & evolving** detection logic directly in Solidity is prohibitively **expensive** and complex.
- Even if you could, it would hardly adapt as smart contracts are too **rigid**
- You might need to protect assets of smart contracts **you don't control**
- Reaction might require **on-chain and off-chain** actions

...by the time you update the contract, the attack has already moved on.

How it works: simple version



How it might/will work



Examples

- Monitor all transactions on a smart contract
- Monitor cross-chain assets (USDC/USDT...)
- Monitor all stablecoins backed by USD/EUR
- Consider broad chain data (eth price, gasprice...)

Why The Graph

- Broad data availability
- High throughput
- Flexibility over time
- Fast indexing & retrieving

What we used

- Substreams Development (substreams-dev)

Use cases and applications

- Detect AML events
- Detect new hacks (classic and distributed)
- Enforce compliance with regulations (**MICAr**)
- Asset protection